

Revised Generation Interconnection System Impact Study Report

For Queue Project Q21-1947

Roanoke Rapids 138 kV
November 26, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1947. The Interconnection Customer proposes a 138.8 MW solar photovoltaic facility interconnecting at the Roanoke Rapids 138 kV substation in the North Carolina area. The request was received on November 30, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 679 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 11 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$669,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
N Tower Tap - Cedar Falls Wi 138.0 kV	North Tower Tap - Haggin Substation 345.0 kV	109.5	470	6.4
Haggin Substation 345-138 kV Autotransformer	N Tower Tap - Haggin Substation 345.0 kV	104.8	543	5.3
Kings Riv - Dg Amp Solar Coldwater 138.0 kV	Mitchell Dam Substation - Skaggs Island Jct 2 345.0 kV Line	103.6	530	5.1
Kings Riv - South Meado 2 138.0 kV L	Mitchell Dam - Skaggs Island Junction 2 345.0 kV line	103.3	484	5.1
Cedar Crk - Yorkana 345.0 kV line	Co Line Rd - Monroe Ne 500.0 kV	101.4	1,494	6.8
Spring Street - Chipola 138.0 kV line	Mitchell Dam - Skaggs Island Junction 2 345.0 kV	101.1	450	7.5
Natchez - Chancello Row 138.0 kV L	Skaggs Island Jct 2 - Mitchell Dam 345.0 kV	101.1	692	6.7
N Tower Tap - Unknown 138.0 kV L	N Tower Tap - Haggin 345.0 kV Line	101.0	675	10.2
Walthall - Unknown 138.0 kV	Haggin - N Tower Tap 345.0 kV	101.0	658	10.2
Caliente - Magnolia 138.0 kV	Cosmar - Ht Rapids 138.0 kV Line	101.0	529	7.7
Hull Street Rd - Harvard 138.0 kV	Mitchell Dam - Skaggs Island Jct 2 345.0 kV	100.5	806	9.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Cedar Falls Wi - N Tower Tap 138.0 kV line	Reconductor line	\$3,507,000	3% (\$115,000)
Haggin 345/138kV TX	Replace transformer with higher-rated unit	\$17,195,000	1% (\$217,000)
Dg Amp Solar Coldwater - Kings Riv 138.0 kV L	Reconductor line	\$1,583,000	1% (\$16,000)
Kings Riv - South Meadow 2 138.0 kV	Reconductor line	\$2,208,000	1% (\$23,000)
Cedar Crk - Yorkana 345.0 kV line	Reconductor line	\$4,962,000	1% (\$44,000)
Spring Street - Chipola 138.0 kV line	Reconductor line	\$3,788,000	2% (\$65,000)
Natchez - Chancello Row 138.0 kV L	Reconductor line	\$1,101,000	1% (\$11,000)
N Tower Tap - Unknown 138.0 kV L	Reconductor line	\$2,041,000	2% (\$32,000)
Walthall - Unknown 138.0 kV L	Reconductor line	\$7,852,000	2% (\$123,000)
Caliente - Magnolia 138.0 kV	Reconductor line	\$1,108,000	1% (\$16,000)
Hull Street Rd - Harvard 138.0 kV	Reconductor line	\$476,000	1% (\$7,000)
	Total allocated to project		\$669,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.